PLC

Timer and Counter Example

Write a logic ladder diagram to implement the following process using a PLC.

1. Start push-Button is used to start the process and a stop push-Button is used to stop the process.

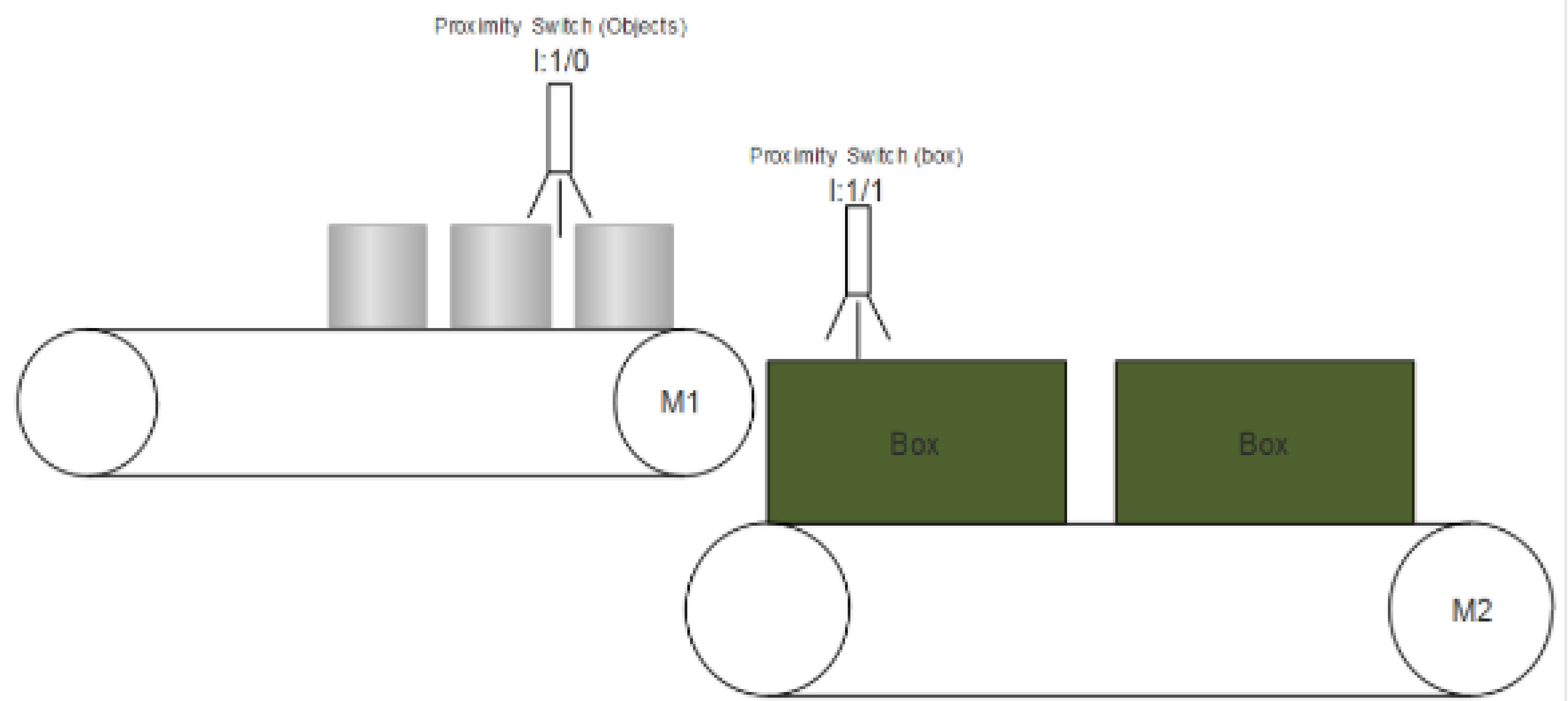
2. A proximity switch is used to detect moving objects on the conveyor belt 1.

3. Another proximity switch is used to detect an empty box on conveyor belt 2.

4. 6 objects should be placed in every box.

5.When number of objects to be packed are detected timer is activated the time between the moving of empty boxes is set to be 2 sec. When time is over, it moves the conveyor bell 2 until next empty box is detected.

Problem Diagram



PLC Program

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List of Inputs and Outputs
 I:1/14      = Start                                 (Input)
 I:1/15      = Stop                                  (Input)
 B3:0/0      = Latching Bit                          (Bit)
 B3:0/1      = Latching Motor 1 bit                  (Bit)
 I:1/0       = Part detection                        (Input)
 I:1:1       = Box detection                         (Input)
 O:2/0       = Conveyor Belt 1                       (Output)
 O:2/1       = Conveyo2 Belt 2                       (Output)
 C5:0 = Part counter                                 (Counter)
 T4:0 = Timer to stop conveyor                       (Timer)
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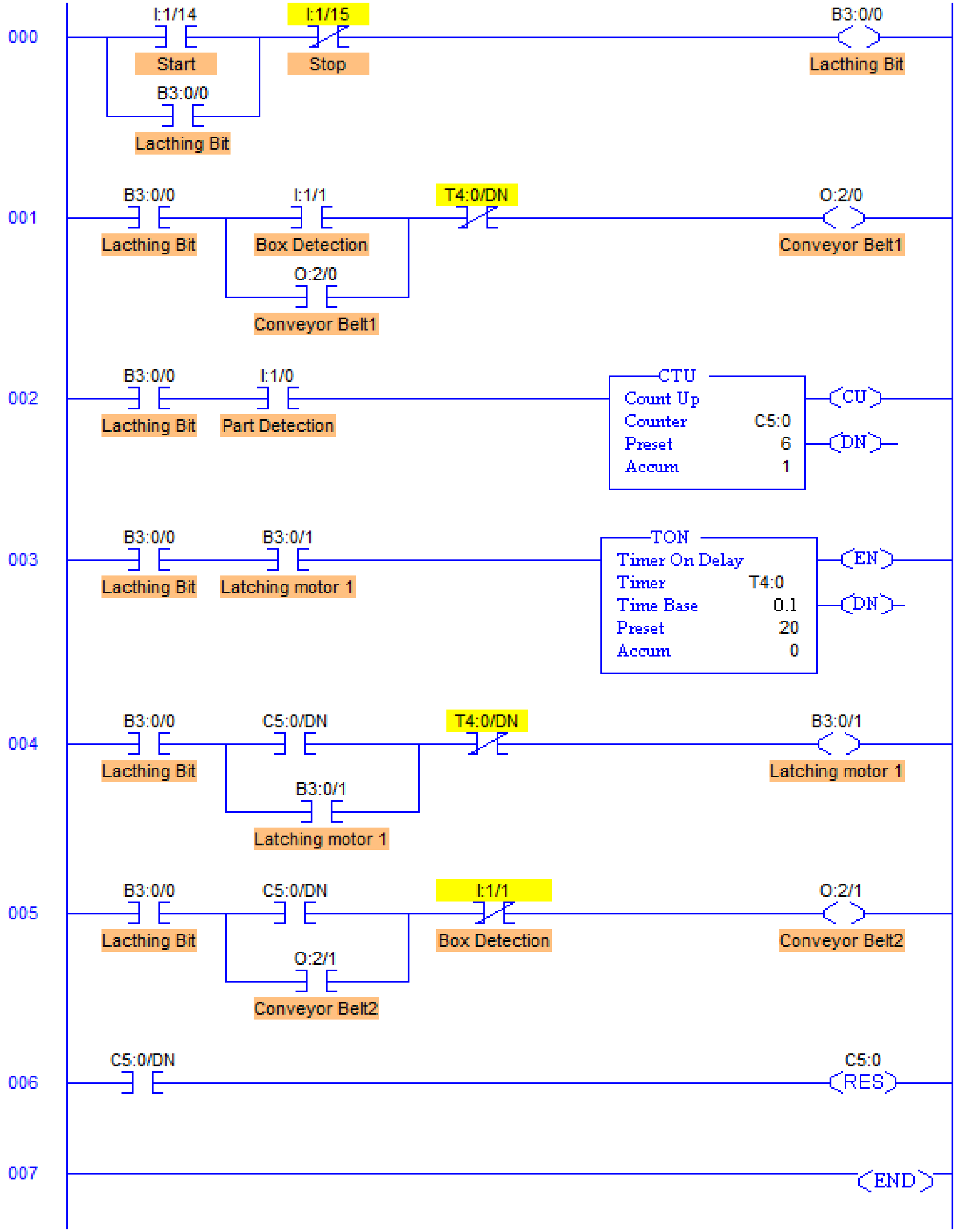

000
I:1/14
Start
B3:0/0
Lacthing Bit
I:1/15
Stop
B3:0/0
Lacthing Bit
001
B3:0/0
Lacthing Bit
I:1/1
Box Detection
O:2/0
Conveyor Belt1
T4:0/DN
O:2/0
Conveyor Belt1
002
B3:0/0
Lacthing Bit
I:1/0
Part Detection
CTU
Count Up
Counter C5:0
Preset 6
Accum 1
CU
DN
003
B3:0/0
Lacthing Bit
B3:0/1
Latching motor 1
TON
Timer On Delay
Timer T4:0
Time Base 0.1
Preset 20
Accum 0
EN
DN
004
B3:0/0
Lacthing Bit
C5:0/DN
B3:0/1
Latching motor 1
T4:0/DN
B3:0/1
Latching motor 1
005
B3:0/0
Lacthing Bit
C5:0/DN
O:2/1
Conveyor Belt2
I:1/1
Box Detection
O:2/1
Conveyor Belt2
006
C5:0/DN
C5:0
RES
007
END

Program Description

- RUNG000 is for master start and stop the process.
- RUNG001 is for controlling Conveyor Belt 1 motor M1 which is operated by Box Detection proximity switch. When the process is started and empty box is detected by Proximity having address of I:1/1, power supply to the motor of Conveyor Belt 1 is given. This drives the Motor M1 moving objects on the conveyor belt 1.
- Every time a part is detected by the I:1/1 Part detector, Counter C5:0 is incremented by 1. Counter has a preset value of 6, counter done bit C5:0/DN is set to 1.
- When C5:0/DN bit is set, timer T4:0 is activated for 2secs which is an assumption that the part will take approximately 2secs to fall into the box after it is detected by the Pat Detector proximity switch I:1/0.
- Counter is immediately reset when C5:0/DN goes true. So one shot is generated by C5:0/DN bit.
- Conveyor Belt 2 motor M2 (O:2/1) is operated when counter value reaches to 6.
- And the process is repeated.

PLC Program to Drive Motors Simultaneously with Interlocking

Problem Description

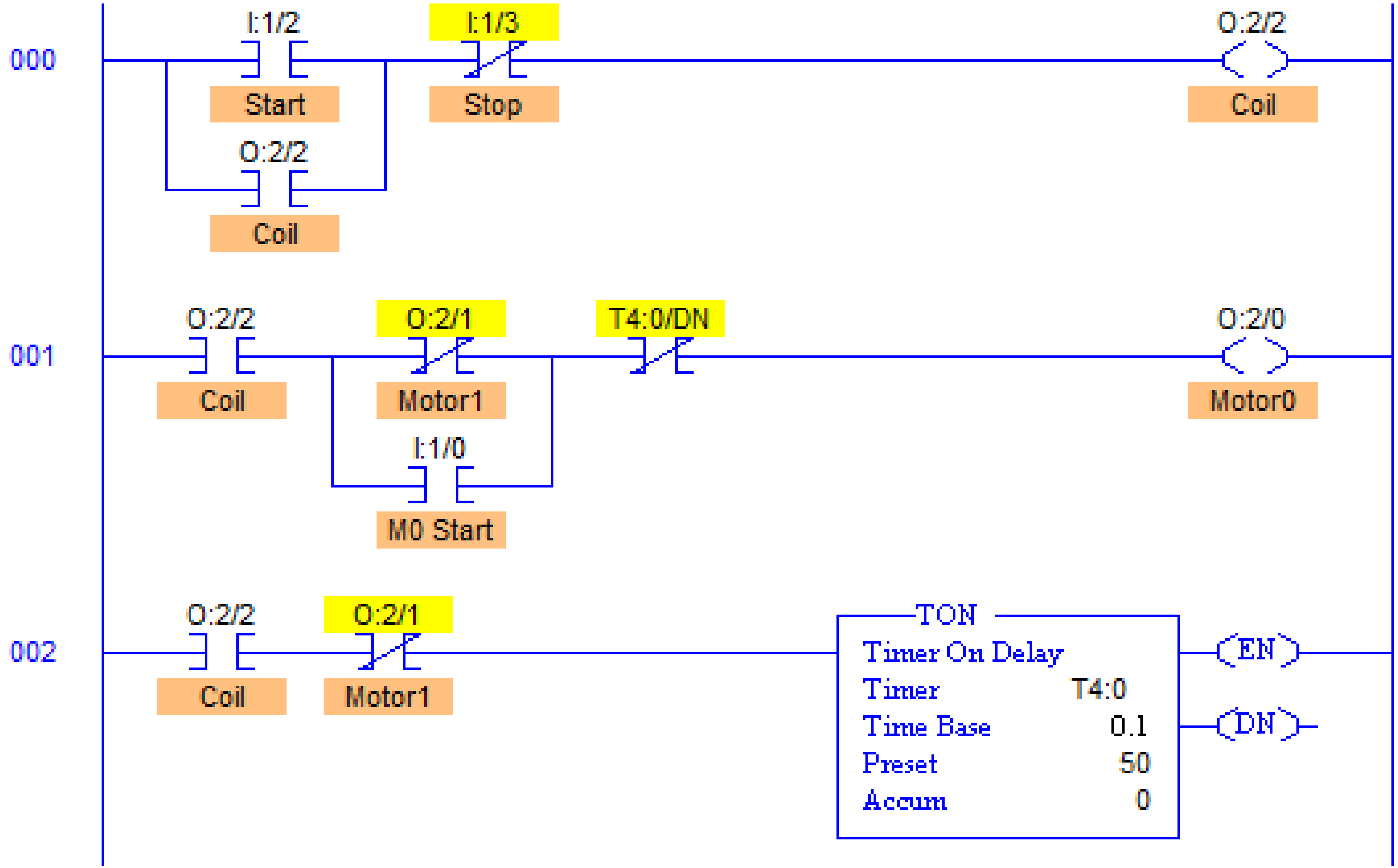
Two Motors are running in a sequence one by one for a particular time. If the start button is pressed Motors run in sequence such that 1st Motor stays ON for 5secs and then 2nd Motor is turned ON and stays ON for 5secs. And the cycle is repeated until it is interrupted. While motors are running in the sequence, if one motor is running and the button of other motor is pressed, then the running Motor should stop and the other motor should run. Implement this logic in PLC using Ladder Diagram programming language.

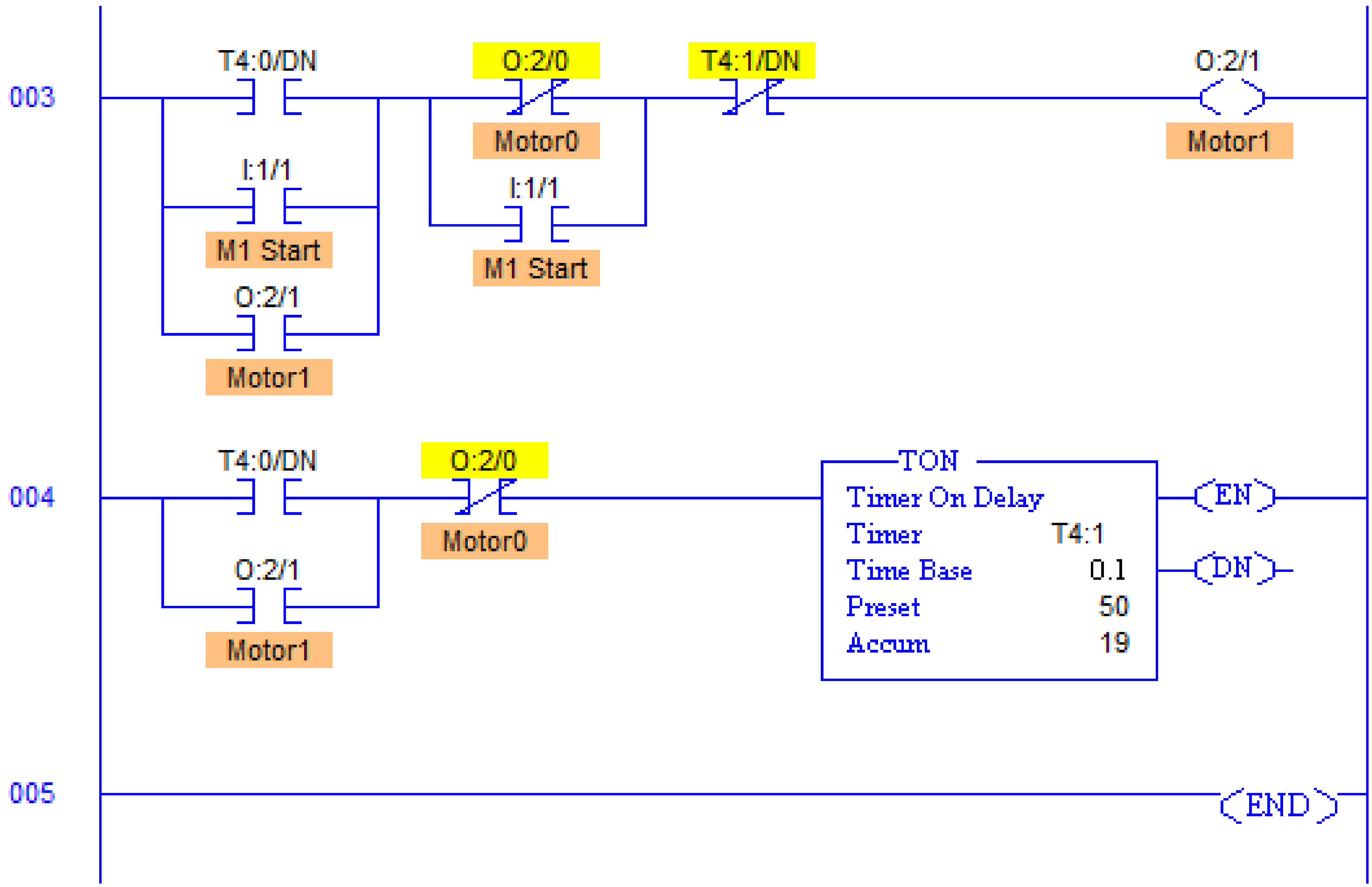
Problem Solution

- Define addresses of motors.
- Use pushbuttons to start and stop the sequential operation.
- Use TON timer to generate a particular time delay.
- Use the enable bit of first timer to energize 2nd motor coil and activate second timer.
- No need to use Reset contact, program will reset the timers itself after completion of each cycles.
- Use Parallel Start Motor contact to start one motor when other motor is running.

PLC Program

- List of Inputs and Outputs
- I:1/0 = Motor 0 Start (Input)
- I:1/1 = Motor 1 Stop (Input)
- I:1/2 = Master Start (Input)
- I:1/3 = Master Stop (Input)
- O:2/2 = Master Coil (Output)
- O:2/0 = Motor 0 (Output)
- T4:0 = Motor 0 Timer (Timer)
- O:2/1 = Motor 1 (Output)
- T4:1 = Motor 1 Timer (Timer)


000
I:1/2
Start
O:2/2
Coil
I:1/3
Stop
O:2/2
Coil
001
O:2/2
Coil
O:2/1
Motor1
I:1/0
M0 Start
T4:0/DN
O:2/0
Motor0
002
O:2/2
Coil
O:2/1
Motor1
TON
Timer On Delay
Timer T4:0
Time Base 0.1
Preset 50
Accum 0
EN
DN


003
T4:0/DN
O:2/0
T4:1/DN
O:2/1
Motor0
Motor1
I:1/1
M1 Start
I:1/1
M1 Start
O:2/1
Motor1
004
T4:0/DN
O:2/0
Motor0
O:2/1
Motor1
TON
Timer On Delay
Timer T4:1
Time Base 0.1
Preset 50
Accum 19
EN
DN
005
END

Program Description

- RUNG000 is Latching coil for Master Start and Stop.
- RUNG001-RUNG002 and RUNG003-RUNG004 are arrangements to operate Motor0 O:2/0 and Motor1 O:2/1 respectively.
- When Start button is pressed momentarily, it latches the coil starting the energizing the output O:2/0, hence turning Motor0 ON.
- Timer T4:0 is used to run Motor0 for 5secs (Less because of simulation purpose).
- Similarly T4:1 is used to run Motor1 for 5secs.

- While Motor0 is running, and M1 Start I:1/1 is pressed, it stops the Motor0 (O:2/0 de-energizes) and Motor1 (O:2/1 energizes) starts and timer t4:1 is activated. And vice-versa.

- XIO contact of each motor’s output is used to turn other motor off by de-energizing other motor’s output which cuts off the current supply in the field.

PLC Example

Traffic lights

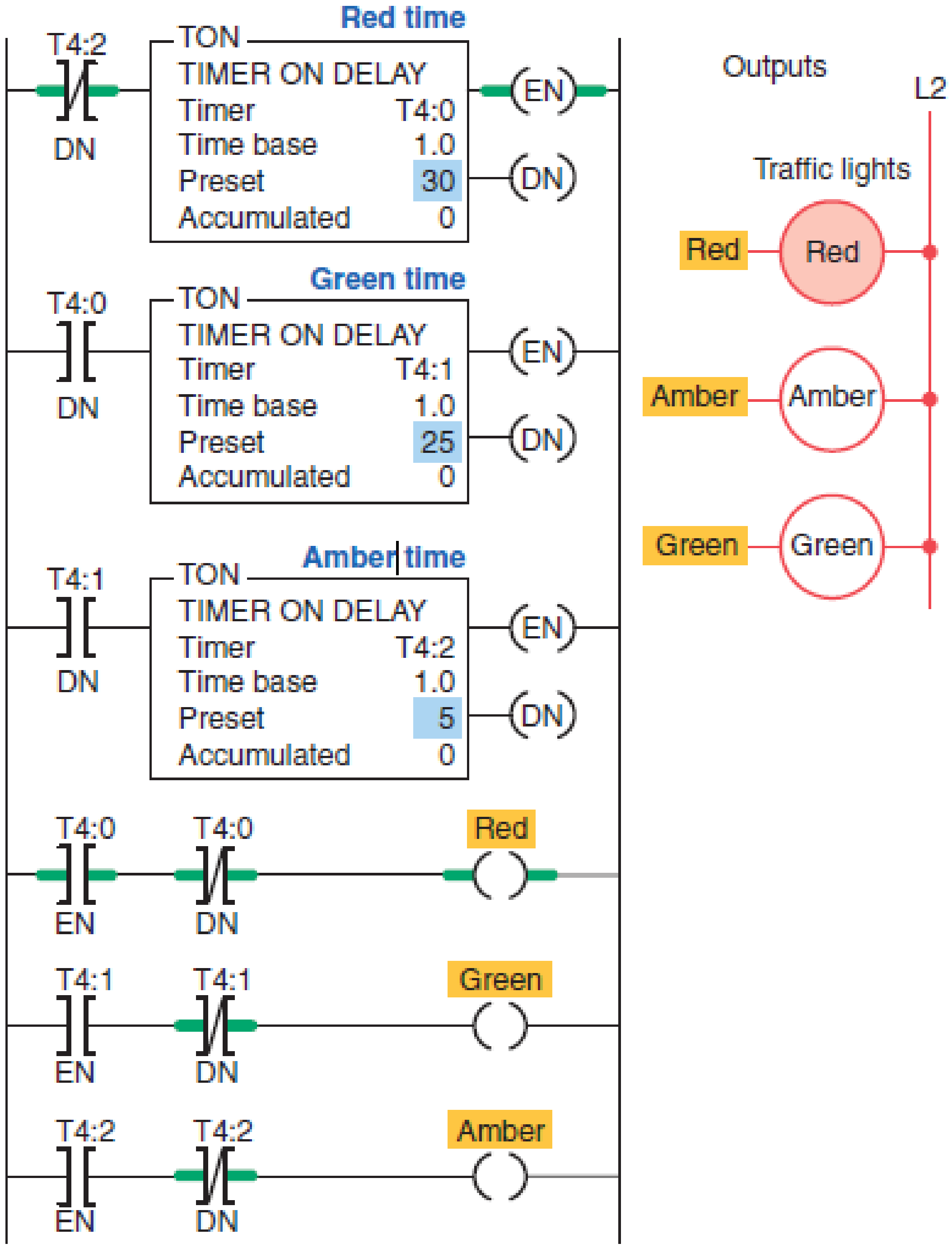
Control of a set of traffic lights in one direction

Operation of the program:

- Transition from red light to green light to amber light is accomplished by the interconnection of the three TON timer instructions.
- The input to timer T4:0 is controlled by the T4:2 done bit.
- The input to timer T4:1 is controlled by the T4:0 done bit.
- The input rung to timer T4:2 is controlled by the T4:1 done bit.

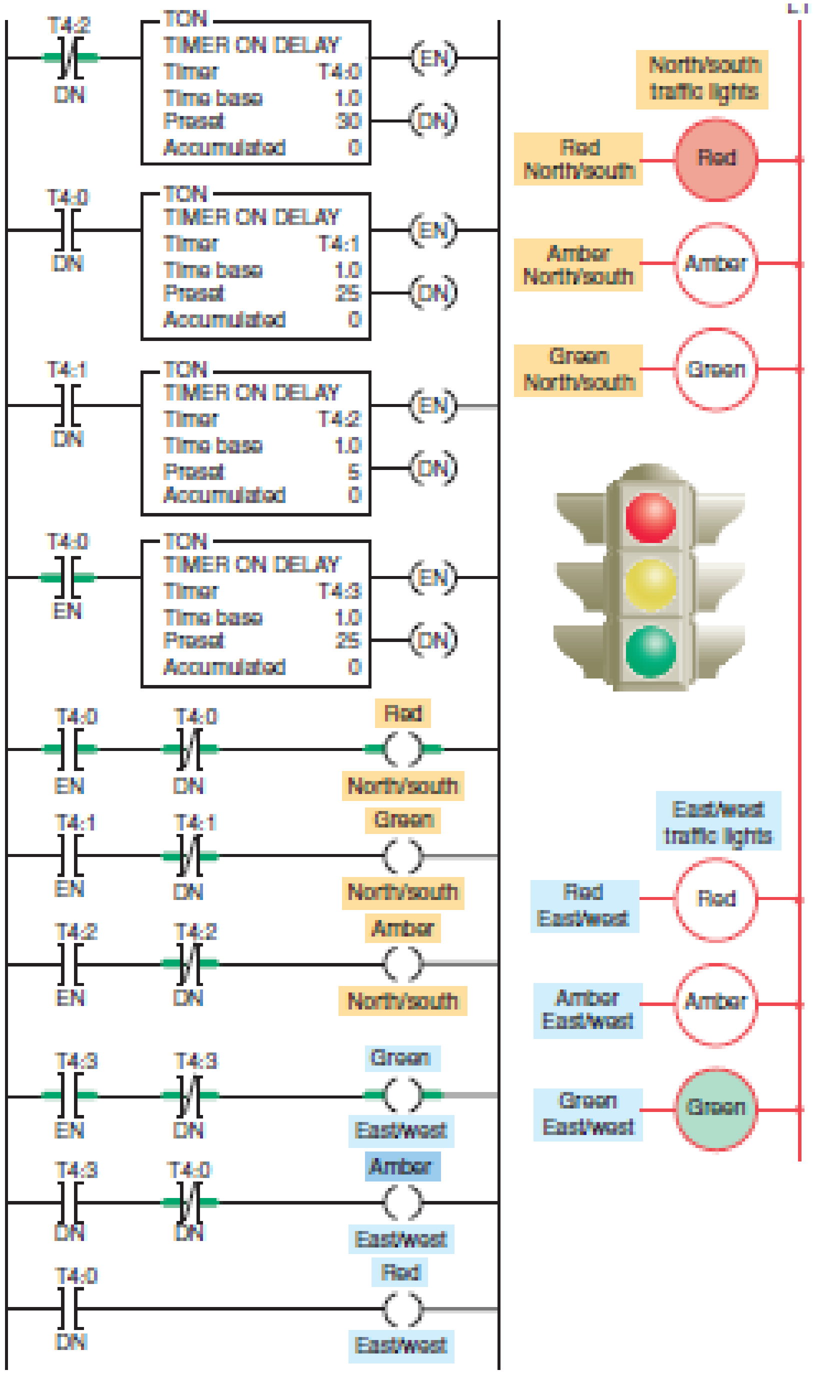
- The timed sequence of the lights is:
- Red—30 s on
- Green—25 s on
- Amber—5 s on

The sequence then repeats itself.



Red time
T4:2
DN
TON
TIMER ON DELAY
Timer T4:0
Time base 1.0
Preset 30
Accumulated 0
EN
DN
Green time
T4:0
DN
TON
TIMER ON DELAY
Timer T4:1
Time base 1.0
Preset 25
Accumulated 0
EN
DN
Amber time
T4:1
DN
TON
TIMER ON DELAY
Timer T4:2
Time base 1.0
Preset 5
Accumulated 0
EN
DN
T4:0 EN
T4:0 DN
Red
T4:1 EN
T4:1 DN
Green
T4:2 EN
T4:2 DN
Amber
Outputs
L2
Traffic lights
Red
Red
Amber
Amber
Green
Green

Control of a set of traffic lights in two direction



L1
T4:2
DN
TON
TIMER ON DELAY
Timer T4:0
Time base 1.0
Preset 30
Accumulated 0
(EN)
(DN)
T4:0
DN
TON
TIMER ON DELAY
Timer T4:1
Time base 1.0
Preset 25
Accumulated 0
(EN)
(DN)
T4:1
DN
TON
TIMER ON DELAY
Timer T4:2
Time base 1.0
Preset 5
Accumulated 0
(EN)
(DN)
T4:0
EN
TON
TIMER ON DELAY
Timer T4:3
Time base 1.0
Preset 25
Accumulated 0
(EN)
(DN)
T4:0 EN
T4:0 DN
Red North/south
T4:1 EN
T4:1 DN
Green North/south
T4:2 EN
T4:2 DN
Amber North/south
T4:3 EN
T4:3 DN
Green East/west
T4:3 DN
T4:0 DN
Amber East/west
T4:0 DN
Red East/west
North/south traffic lights
Red North/south
Red
Amber North/south
Amber
Green North/south
Green
East/west traffic lights
Red East/west
Red
Amber East/west
Amber
Green East/west
Green